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Bottom load water dispenser with an elongated dispensing area

Abstract

A bottom loaded water dispenser having a water dispensing area that can be elongated to accommodate any size refillable bottle or other container, including large or wide based bottles or refillable containers. The bottom loaded water dispenser accommodates a standard bottom loaded water bottle. The bottom loaded water dispenser has a door that fits into the recess of the dispenser housing that does not interfere with bottom loaded water bottle and allows the drip tray to be adjusted to any height under the dispensing nozzle.

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Background/Summary

BACKGROUND

(1) Water dispensers have been available for quite some time. Early versions of such dispensers were “top loaded” dispensers, which are still in use today. The top loaded configuration is cumbersome and requires a person to lift a heavy bottle of water, flip it and position it in the top of the water dispenser. Because of this, often, a person changing the water bottle can spill the water; can be injured or considerably strain a muscle, and when the heavy bottle is not properly loaded because it is difficult to control, the flipped bottle may damage the water dispenser.

(2) “Bottom loaded” dispensers were developed to address many of the issues caused by top loaded dispensers. Over time, due to standardization of components and height requirements, legal

requirements and to reduce costs, bottom loaded dispensers are now designed and built to meet certain, pre-defined specifications. For instance, the 5-gallon water bottle and its defined shape and size have become an industry standard. So, the bottom loaded water cooler needs to accommodate its shape and size. Modern water coolers must comply with the American Disabilities Act (“ADA”) and meet certain specifications, such as the controls on the front of the unit must be within the range of 15 inches to 48 inches from the floor to accommodate someone in a wheelchair that wants to use the water dispenser. Further, to make it more efficient and cost-effective to transport multiple water dispensers, water dispensers are sized to get to get the maximum number of water dispensers in a shipping container. Because of the limitations a bottom loaded water dispenser needs to satisfy, a bottom loaded water dispenser on the market today typically has a height in the 48 inch to 52 inch range.

(3) In prior designs, in addition to the top height constraints on a bottom loaded water dispenser discussed above, the drip tray was set at a height on the water cooler higher than the top most height of the 5-gallon water bottle to allow the water bottle to be easily moved in and out for replacement. With these prior designs, the top height limitation of the cooler itself and the minimum height that the drip tray can be placed on the cooler, due to the standardized 5-gallon water bottle height, created a “maximum dispensing height” on a bottom loaded water dispenser for a bottle or container that can filled by the dispenser. Such limitations are acknowledged and commented on in US2014/0131385, par. 0035.

(4) However, over time, people have become more health conscious and with this, they drink more water during the course of the day. As a result, people are using taller refillable bottles and containers than they did in the past. These refillable taller bottles and containers do not fit in the present “maximum dispensing height” between the dispensing nozzles and the drip tray of a bottom loaded water dispenser. Re-designing the water dispenser to keep the drip tray on the door while moving the position of drip tray down on the door to elongate the dispensing height and create a larger “maximum dispensing height” is not practical because it would create an unaesthetically pleasing seam where the door engages the water dispenser housing, since the housing has to have an opening tall enough to insert and remove a 5-gallon water bottle.

(5) Accordingly, there is a need for an aesthetically pleasing bottom loaded water dispenser that can accommodate refillable bottles and containers of varying heights within the present build constraints of a bottom loaded water dispenser.

SUMMARY

(6) According to one aspect of the present invention, a bottom loaded water dispenser with an elongated dispensing area, having a housing which includes a dispensing nozzle, a front side and a lower portion and where the nozzle has an end tip and where a recess is formed in the front side of the housing behind the dispensing nozzle and below the dispensing nozzle end tip and where an opening is formed in the lower portion of the housing to receive a water bottle and also having a door connected to the housing having a drip tray; where the door is moveable between an open and closed position and where the recess formed in the front side of the housing is approximately equal to the thickness of the door, such that in the closed position, the door fits in the recess and the top of the door is below the end tip of the dispensing nozzle. The bottom loaded water dispenser of the present invention may also have an adjustable drip tray. The bottom loaded water dispenser of the present invention may also include a lower lip portion that defines the top of the formed recess and abuts the top of the door in the closed position, and the lower lip portion may be separate from the other housing components. The bottom loaded water dispenser of the present invention may have a door that connects to the housing with hinges, and the door may also be curved. The bottom loaded water dispenser of the present invention may also include an access panel in the housing.

Description

DRAWINGS

- (1) Objects, features, and advantages of the present invention will become apparent upon reading the following description in conjunction with the drawing figures, in which:
- (2) FIG. 1A is a perspective view of a bottom loaded water dispenser of the present invention with a door closed;
- (3) FIG. 1B is a front view of a bottom loaded water dispenser of the present invention with a door closed;
- (4) FIG. 2A is a perspective view of a bottom loaded water dispenser of the present invention with the door open;
- (5) FIG. 2B is a front view of a bottom loaded water dispenser of the present invention with the door open;
- (6) FIG. 3 is a partial perspective view of the top of the bottom loaded water dispenser of the present invention;
- (7) FIG. 4 is a partial side view of the top of the bottom loaded water dispenser of the present invention;
- (8) FIG. 5 is a perspective view of a bottom loaded water dispenser of the present invention with the door open, and illustrating a water bottle being removed;
- (9) FIG. 6A is a partial perspective view of the top of an embodiment of a bottom loaded water dispenser of the present invention having an access panel;
- (10) FIG. 6B is a partial perspective view of the top of the embodiment of the bottom loaded water dispenser of the present invention having an access panel with the access panel removed;
- (11) FIG. 7A is a partial perspective view of the top of an embodiment of a bottom loaded water dispenser of the present invention where the door of the water dispenser is curved; and
- (12) FIG. 7B is another perspective view of the top of the embodiment of a bottom loaded water dispenser of the present invention where the door of the water dispenser is curved.

DESCRIPTION

- (13) Referring to FIGS. 1A, 1B, an embodiment of a bottom loaded water dispenser **10** of the present invention is depicted. The water dispenser **10** includes a housing **12** and a door **14**. The housing **12** contains the operating components **16** (FIG. 6B) of the water dispenser **10** in an upper interior space **18**. The housing **12** also defines a lower interior space **20** (FIGS. 2A, 2B), which accommodates a water bottle **22** when a water bottle is loaded into the water dispenser **10**. The lower interior space **20** has a height, width and depth large enough to accept an inserted water bottle **22**. The housing **12** also includes a water dispensing housing **24** that houses water dispensing buttons **26** and one or more dispensing nozzles **28**. The housing **12** also includes a lower lip portion **30** that is set along the base of the water dispensing housing **24**. In differing embodiments, the lower lip portion **30** can be formed integral with the water dispensing housing **24**, be formed as a separate, stand-alone component or be formed integral with another component of the water dispenser **10** of the present invention. In this embodiment, the dispensing nozzles **28** each have an end tip **29**, which is the lowest point of the dispensing nozzle **28** relative to the ground. The dispensing buttons **26** interact with the operating components **16** to dispense water to a user through the dispensing nozzle **28** when the buttons **26** are pushed.
- (14) In this embodiment, the door **14** of the water dispenser **10** is separate from the housing **12** and is connected to the housing **12**. The door **14** is connected to the housing **12** with hinges (not shown) in this embodiment. Referring to FIGS. 2A-B, 3 and 4, in a preferred embodiment, the lower lip portion **30** defines the top of a recess **40** so that the door **14**, when engaged with the front of the housing **12** in a closed position, fits in the recess **40** defined on the front of the housing **12** under the lower lip portion **30**. The lower lip portion **30**, and the recess **40** that it forms the top of, are approximately equal to the thickness of the door **14** so that the door **14**, when closed, fits cleanly within the recess **40** and does not extend out from underneath the lower lip portion **30**. Further, the

lower lip portion **30** is positioned at a position lower than the end tip **29** of a dispensing nozzle **28**. Placing the lower lip portion **30** at this position on the dispenser **10** keeps the dispensing nozzle **28** from interfering with the opening and closing of the door **14**.

(15) Referring again to FIGS. **1A-1B** and **2A-2B**, the door **14** and the recess **40** are designed so that a water bottle **22** can fit entirely within the interior space **20** without interfering with closing the door **14**. The door **14** includes a base portion **42**, a front bottom portion **44** and a drip tray **46**. In the embodiment depicted, the base portion **42** extends the entire length of the door **14** behind the front bottom portion **44**. In other embodiments, the base portion **42** might not extend the entire length of the door **14**, and the front bottom portion **44** would engage with the housing **12** when the door **14** is closed.

(16) Since the drip tray **46** is connected to the door **14**, and the door **14** is the full length of the recess **40** in the housing **12**, the drip tray **46** can be placed anywhere along the face of the door **14** to create an elongated dispensing height **50** (FIG. **1B**) that can accommodate any size refillable bottle or containers, including very tall bottles or containers. Thus, a water dispenser **10** of the present invention can have a significantly large “maximum dispensing height” compared to prior water coolers, and the “maximum dispensing height”, can, in theory, be set to span from the bottom of the end tip **29** of the dispensing nozzle **28** all the way to the bottom of the housing **12**, if a user wanted a dispensing height **50** that large. As such, the elongated dispensing height **50** in preferred embodiments of the present invention is much larger than the present maximum dispensing height used in the industry. In particular, the drip tray height, and correlated dispensing height **50**, in the present invention are not limited by the seam formed at the top of the interior space **20**, which has to be tall enough to accommodate a water bottle **20**.

(17) Referring to FIGS. **2A**, **2B** and **5**, in operation, a water bottle **22** is inserted into the interior space **20** of the housing **12**. The water bottle **22** is connected to the operating components **16**. The door **14** is then closed, concealing the water bottle **22** behind it (FIGS. **1A**, **1B**) and since the entire door **14** fits into the recess **40** of the housing **12**, there is no seam showing at the edge of the interior space **20**. The adjustable tray **46** has been set at the desired height to accommodate whatever size bottles need to be accommodated during filling. With the door **14** closed, a user of the water dispenser **10** may insert a large refillable bottle or other container into the elongated dispensing height **50**, underneath the nozzle **28**, to fill the user's bottle or container with water.

(18) Referring to FIG. **5**, when the water bottle **22** is empty, the user of the dispenser **10** opens the door **14**, disconnects the bottle **22** from operating components **16** and removes the water bottle **22** from the interior space **20**. The user then inserts a new water bottle **22**, connects it to the operating components **16** and closes the door **14**. Again, in this closed configuration, because of the size of the door **14** and its engagement with the recess **40**, no seam shows at the edge of the interior space **20**.

(19) Referring to FIGS. **6A-6B**, another embodiment of the water dispenser **10** of the present invention is depicted. In this embodiment, the housing **12** has an access panel **60** in the recess area **40** of the housing **12**. In this embodiment, the access panel **60** allows a technician to remove the access panel **60** (FIG. **6B**) to easily access the operating components **16** if they need to be serviced or repaired. With present coolers that do not have a door **14**, it is very difficult, if not impossible, for a service technician to access the operating components **16**. The access panel **60** also serves the purpose of preventing the water dispenser user from seeing and touching the operating components **16**, which could cause injury. Further, the door **14** allows access panel **60** to be used without ruining the aesthetic of the water dispenser **10** because when the door **14** is in the closed position, the access panel **60** cannot be seen by the water dispenser user.

(20) Referring to FIGS. **7A-7B**, a further embodiment of the water dispenser **10** of the present invention is depicted. In this embodiment, the door **14** is curved towards the back of the housing **12** as indicated at **52** (FIG. **7B**). This curvature maximizes the distance from the end tip **29** of the nozzle **28** to the door **14**, allowing a dispenser user to use a variety of containers having a large

width, including those with a wide top or a wide base.

(21) Although certain embodiments and features of a bottom loaded water dispensing device have been described herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all embodiments of the teachings of the disclosure that fairly fall within the scope of permissible equivalents.

Claims

1. A bottom loaded water dispenser with an elongated dispensing area, comprising: a housing having a dispensing nozzle, a front side and a lower portion; wherein the nozzle has an end tip and wherein a recess is formed in the front side of the housing behind the dispensing nozzle and below the dispensing nozzle end tip and wherein an opening is formed in the lower portion of the housing to receive a water bottle; a door connected to the housing having a drip tray; wherein the door is moveable between an open and closed position and wherein the recess formed in the front side of the housing is approximately equal to the thickness of the door, such that in the closed position, the door fits in the recess and the top of the door is below the end tip of the dispensing nozzle.
2. The bottom loaded water dispenser of claim 1, wherein the drip tray is adjustable.
3. The bottom loaded water dispenser of claim 1, wherein the housing, further comprises a lower lip portion that defines the top of the formed recess and abuts the top of the door in the closed position.
4. The bottom loaded water dispenser of claim 3, wherein the housing is comprised of several components and the lower lip portion is a component that is separate from the other component.
5. The bottom loaded water dispenser of claim 1, wherein the door is connected to the housing with hinges.
6. The bottom loaded water dispenser of claim 1, wherein the housing includes an access panel.
7. The bottom loaded water dispenser of claim 1, wherein the door is curved.
8. A bottom loaded water dispenser with an elongated dispensing area, comprising: a housing having a dispensing nozzle, a front side and a lower portion; wherein the nozzle has an end tip and wherein a recess is formed in the front side of the housing behind the dispensing nozzle and below the dispensing nozzle end tip and wherein an opening is formed in the lower portion of the housing to receive a water bottle; a door connected to the housing having an adjustable drip tray; wherein the door is moveable between an open and closed position and wherein the recess formed in the front side of the housing is approximately equal to the thickness of the door, such that in the closed position, the door fits in the recess and the top of the door is below the end tip of the dispensing nozzle.
9. The bottom loaded water dispenser of claim 8, wherein the housing, further comprises a lower lip portion that defines the top of the formed recess and abuts the top of the door in the closed position.
10. The bottom loaded water dispenser of claim 9, wherein the housing is comprised of several components and the lower lip portion is a component that is separate from the other component.
11. The bottom loaded water dispenser of claim 8, wherein the door is connected to the housing with hinges.
12. The bottom loaded water dispenser of claim 8, wherein the housing includes an access panel.
13. The bottom loaded water dispenser of claim 8, wherein the door is curved.
14. A bottom loaded water dispenser with an elongated dispensing area, comprising: a housing having a dispensing nozzle, a lower lip portion, a front side and a lower portion; wherein the nozzle has an end tip and wherein a recess is formed in the front side of the housing behind the dispensing nozzle and below the dispensing nozzle end tip and the lower lip portion defines the top of the recess and wherein an opening is formed in the lower portion of the housing to receive a water bottle; a door connected to the housing having an adjustable drip tray; wherein the door is

moveable between an open and closed position and wherein the recess formed in the front side of the housing is approximately equal to the thickness of the door, such that in the closed position, the door fits in the recess and the top of the door abuts the lower lip portion and is below the end tip of the dispensing nozzle.

15. The bottom loaded water dispenser of claim 14, wherein the housing is comprised of several components and the lower lip portion is a component that is separate from the other component.

16. The bottom loaded water dispenser of claim 14, wherein the door is connected to the housing with hinges.

17. The bottom loaded water dispenser of claim 14, wherein the housing includes an access panel.

18. The bottom loaded water dispenser of claim 14, wherein the door is curved.
